

2 In the fruit fly, *Drosophila melanogaster*, two different genes control body colour and eye colour.

- **G/g** are alleles of the body colour gene.
- **G** results in grey body, **g** results in black body.
- **R/r** are alleles of the eye colour gene.
- **R** results in red eyes, **r** results in brown eyes.

Each gene is autosomal.

A dihybrid cross was carried out using a fly with a grey body and red eyes crossed with a fly with a black body and brown eyes. Both parents were homozygous for both genes. The offspring from the F1 generation were crossed to obtain the F2 offspring.

- (a) A statistical test showed that the results of the cross were significantly different from those expected.

State the name of the statistical test used **and** state the expected phenotypic ratio for the F2 generation.

statistical test

expected ratio

[2]

- (b) A test cross can be carried out in order to identify flies from an F₂ generation that are heterozygous for both genes.

Draw a genetic diagram to show how a test cross between a heterozygous grey-bodied, red-eyed F₂ fly and a fly with a black body and brown eyes can produce four different offspring phenotypes.

Use the symbols **G/g** and **R/r**.

- (c) The results of the test cross in (b) are shown in Table 2.1. These results are significantly different from the expected results.

Table 2.1

phenotypes of offspring of test cross	number of individuals
grey body, red eyes	123
grey body, brown eyes	7
black body, red eyes	6
black body, brown eyes	132

Describe how these results are different from the expected results **and** explain why they are different.

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[Total: 11]